Technical Report — ParlTopic: Topic Classification for Swiss Parliaments with Apertus

- **Challenge:** 2 — OpenParlData: Topic classification for Swiss parliaments with Apertus
- **Event:** Hack Apertus Liebefeld, 18–19 October 2026
- **Team:** Token Tell — Felipe Wüthrich, Josip Jukic
- **Repository:** GitHub · **Experiment log:** docs/experiments.md · **Benchmark notes:** docs/benchmark.md
- **Demo:** Streamlit (`make run`: Docker, offline replays; live with an API key) · API (`make api`) · CLI (`parltopic classify`) · pitch and demo script

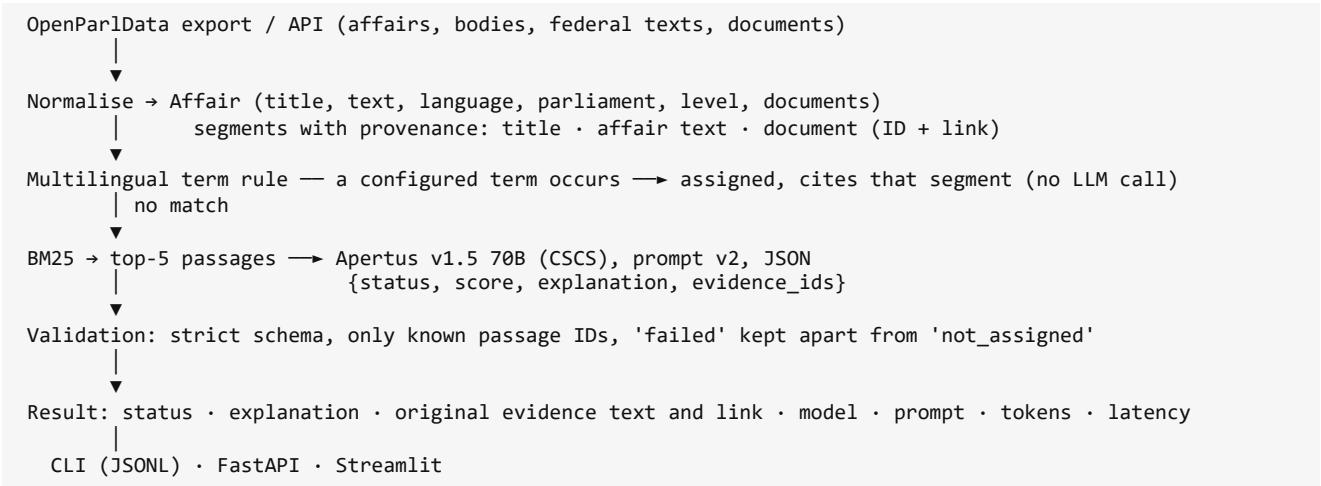
1. Summary

OpenParlData.ch holds more than 300,000 affairs from 90 Swiss parliaments in German, French, Italian and some Romansh, but they can only be searched by full text. A question such as *"What did Swiss parliaments decide on digitalisation?"* means guessing the wording in every language. ParlTopic is an open classification service on **Apertus v1.5 70B**: a user defines a topic in a short YAML file (name, description, terms, examples), and the service decides for each affair whether it belongs to that topic, citing the passage that justifies the decision. New topics need no training.

We validated it on the challenge's real use case: Digital Administration Switzerland (DVS) identifies digitalisation affairs by hand, based on the PolitMonitor editorial tag (3,433 tagged affairs from 35 parliaments). On the 1,200 affairs of 2024–2026 (one third tagged), **Apertus finds 88 % of the tagged affairs at 88 % precision (F1 0.876); keyword search over the same multilingual terms finds 55 % (F1 0.674)**. F1 is 0.90 in German and 0.85 in French and Italian, and 27 of the 30 affairs on the current DVS list are found. Three further topics written once from their name only (migration, health, transport) reach F1 0.78–0.87 against the Federal Assembly's official subject areas, where keyword search reaches 0.55–0.72. Because only 2.3 % of all affairs are digitalisation affairs, about one in four suggestions over the full corpus is PolitMonitor-tagged; keyword search has the same ratio at 63 % of our recall. ParlTopic is therefore a suggestion list for the editors, not an automatic label.

Three findings shaped the system. **The topic definition is the cheapest lever:** an explicit inclusion rule ("side aspects count, a precise title is enough") raised dev F1 from 0.72 to 0.81 without extra cost. **Apertus' own score is not a confidence:** 46 of 64 missed positives had a score ≤ 0.1, so we route by a deterministic multilingual term rule instead of a score threshold (78 of 450 affairs need no model call). **Two people agree on an Année Politique Suisse chapter only 55 % of the time** (κ 0.51, 20 affairs), so a fixed scheme is hard even for humans; user-defined topics are the stronger offer.

2. Architecture



1. **Ingestion.** `parltopic download-data` fetches the bulk exports (`affairs.ndjson.gz`, 326,396 affairs; `bodies.ndjson.gz`) and the hackathon label files. Federal affairs get their submitted text and reasoning in the chosen language; the Federal Council's answer is left out so the executive's framing does not decide the topic. `parltopic enrich-documents` attaches the extracted text of an affair's documents from `/v1/affairs/{id}/docs` (cached).

2. **Segmentation.** Title, affair text and each document become separate segments with stable IDs, so evidence always points to its source field and document.
3. **Term rule.** If the topic name or a configured term occurs (prefix match for words such as *digital/digitale/digitalizzazione*, whole-token match for acronyms such as *NFT*), the affair is assigned with that segment as evidence. Terms inside attached documents can be excluded.
4. **Apertus.** All other affairs go to Apertus with the topic definition, in- and out-of-scope examples and the five passages that BM25 ranks highest against the topic's terms.
5. **Validation.** The answer must match a strict schema; `assigned` and `uncertain` must cite at least one passage; unknown passage IDs are rejected. Technical errors become `failed`, never a negative label. `uncertain` and `not_evaluable` are flagged for review.
6. **Topic policy.** Each topic file can set its prompt version and switch the term rule off. Curated topics with vetted terms (digitalisation) use the rule and prompt v2; topics written in a minute use the generic prompt v1 without the rule (5.3).
7. **Optional stages,** measured and off by default: a second look with Apertus reasoning for rejected affairs (`parltopic rethink`), retrieved few-shot examples, title-only rules.
8. **Human review.** Every suggestion can be confirmed or rejected, with an optional corrected status and note, in the Streamlit demo, both for a single affair and for each hit of a topic search. The review is stored next to the unchanged suggestion and its exact prediction ID; a review never overwrites or deletes the suggestion.
9. **Integration into OpenParlData.** `parltopic export-opd` turns results into rows for a PostgreSQL table `affair_topics` (status, explanation, evidence text and link, model, prompt and topic version, review columns), keyed by affair, topic, topic version and run, so repeated imports are idempotent and old topic versions remain available. A nightly Apache Hop pipeline (table input → `parltopic` → table output) classifies new affairs; the FastAPI service is the REST alternative (docs/integration.md).

3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B`, temperature 0, on the CSCS inference endpoint (OpenAI-compatible, `response_format: json_object`), retries with exponential backoff. The model is set by `LLM_NAME`; nothing else is hard-wired.
- **Role:** Apertus is the semantic classifier and writes the explanation; it chooses the evidence passages by ID, so every cited text is original source text. It also translated the French and Italian cantonal titles into German for our human labelling page. No other model is used for classification or evaluation; no proprietary model is called at runtime.
- **8B vs. 70B (dev, same prompt):** 8B reaches F1 0.43 (recall 0.28) against 0.80 for 70B. It rejects most affairs whose digital aspect is not in the title, and in 169 of 450 answers it returned malformed evidence lists. 70B is the only viable size for this task.
- **Observed repeatability:** two identical 70B runs change 0 of 450 labels. This is evidence for repeatability on that sample, not a guarantee of deterministic serving.
- **Scores are not confidences:** 46 of the 64 missed positives of the first run had a score ≤ 0.1 and only 16 of 450 answers lay between 0.1 and 0.9. A "send low-confidence cases to a second model" cascade cannot find these errors.
- **Reasoning:** Apertus only reasons before answering when the assistant turn is prefilled with `<|inner_prefix|>`; the answer follows `<|inner_suffix|>`. Re-checking rejected affairs this way raised dev F1 from 0.797 to 0.866 with the first prompt, but adds nothing once prompt v2 and the term rule are in place (macro-F1 −0.0015, 95 % CI [−0.015, +0.012]) at 3× the input tokens. It stays optional.

4. Data and evaluation protocol

- **Sources:** OpenParlData bulk exports and the hackathon label directory, snapshot of 18 October 2026, CC BY 4.0.
- **E1 digitalisation (main benchmark):** PolitMonitor-tagged affairs are positives. Negatives are untagged affairs of the same parliaments, period and affair types, sampled per parliament in proportion to its positives so the parliament alone does not reveal the label. **Dev:** 2018–2023, 150 + 300 affairs, used for all design decisions. **Test:** 2024–2026, 400 + 800 affairs (DE 670, FR 325, IT 205; federal 585, cantonal 609, municipal 6). Federal affairs are assigned one of DE/FR/IT at random, so the official translations measure each language on comparable content. Seed 42; `parltopic build-eval` reproduces the files byte for byte.

- **Label caveat:** untagged affairs are positive-unlabelled proxies. Our error analysis found digitalisation affairs among the "false positives" (e.g. *Digital Services Act per la Svizzera*, the SIPAC IT system, the telecom surveillance ordinance), so precision may understate quality.
- **Protocol:** prompts and topic definitions were tuned on dev only. The test split was first run once with predeclared expectations; since then it counts as opened validation, and the frozen configuration was run on it once more with expectations declared beforehand (docs/experiments.md). Paired comparisons use a 10,000-sample bootstrap.
- **Further checks:** E2 — 500 federal affairs with the Federal Assembly's subject areas, used as ground truth for three ad-hoc topics; E3 — 20 affairs labelled blind by both team members with APS chapter and topic; E4 — the 30 affairs on the DVS list on 18 October 2026.

5. Results

5.1 Digitalisation, test split (1,200 affairs, 2024–2026)

Configuration	Precision	Recall	F1	Macro-F1	F1 DE / FR / IT	Tokens in / out per affair
Keyword search, same multilingual terms	0.870	0.550	0.674	0.776	0.64 / 0.68 / 0.76	0 / 0
Apertus 70B, topic v3, first prompt	0.908	0.797	0.849	0.890	0.86 / 0.82 / 0.86	855 / 87
… + reasoning re-check of rejections	0.859	0.885	0.872	0.904	0.88 / 0.87 / 0.85	1,434 / 635
Frozen: term rule + Apertus 70B, prompt v2, topic v4	**0.875**	**0.877**	**0.876**	**0.907**	0.90 / 0.85 / 0.85	1,102 / 80

By level (frozen configuration): federal F1 0.854, cantonal 0.898. Municipal affairs are too few to report (6). 30 affairs on the DVS list: 27 found (keyword search: 19). The frozen run met two of its three predeclared expectations (F1 ≥ 0.84, DVS ≥ 27); the language spread of 0.051 narrowly misses the 0.05 bar. Decisive coverage is 1,193/1,200 (99.4 %): 4 failed, 2 not evaluable and 1 uncertain. Quality metrics use the decisive cases.

Precision at the real base rate. The sample is one third positive; in the corpus 2.3 % of the eligible affairs are tagged. Re-weighting the false-positive rate to that base rate gives the share of suggestions that PolitMonitor tagged:

Configuration	Recall	Precision at 2.3 % base rate
Keyword search	0.55	0.24
Apertus, topic v3, first prompt	0.80	0.32
Apertus + reasoning re-check	0.89	0.22
Frozen	**0.88**	**0.25**

All variants, keyword search included, return about three untagged affairs per tagged one; ParlTopic keeps that ratio at 1.6 times the recall. Part of the untagged suggestions are label gaps (4.1), so the true share is higher, but the editors still have to review the list.

5.2 How we got there (dev split, 450 affairs)

Step	P	R	F1
Keyword search	0.936	0.487	0.640
Apertus 70B, topic definition v1	0.966	0.573	0.720
+ inclusion rule in the definition (v2)	0.952	0.685	0.797
+ side aspects count (v3)	0.955	0.709	0.814
+ term rule and prompt v2 (hybrid-v3, main)	0.932	0.820	0.872
+ topic v4 (frozen)	**0.933**	**0.839**	**0.883**
Rejected: 8B instead of 70B	0.951	0.275	0.426
Rejected: 3 + 3 retrieved labelled examples	0.952	0.673	0.789
Not significant: document text for title-only affairs	0.914	0.859	0.886
Not significant: reasoning re-check on the frozen configuration	0.908	0.859	0.883

The frozen configuration's dev macro-F1 is 0.915; the documents and reasoning variants differ by +0.0009 and −0.0015 with 95 % intervals spanning zero. A title-only TF-IDF classifier trained on 2018–2021 reaches macro-F1 0.82–0.86, below Apertus (Part B of the experiment log).

5.3 Ad-hoc topics beyond digitalisation

Each topic was written once from its name (a two-line description and about 15 DE/FR/IT terms, `configs/topics/adhoc/`) and scored against the Federal Assembly's subject area on 500 federal affairs from 2024–2026.

Topic	Positives	Keyword search	Frozen digitalisation policy	Same, term rule off	**Generic prompt v1, no term rule (final)**
Migration	63	0.717	0.762	0.757	**0.874**
Health	83	0.571	0.656	0.667	**0.805**
Transport	55	0.545	0.642	0.690	**0.784**

The policy tuned for digitalisation does not transfer. Prompt v2 names digital anchors (software, ERP, online platforms) and makes Apertus strict on other topics (precision 0.94–1.00, recall 0.52–0.62), and quickly chosen terms such as *Spital* or *Strasse* make the term rule fire on unrelated affairs (health precision 0.61). Each topic file therefore sets its own policy (`prompt_version`, `keyword_rule`): curated topics with vetted terms use the term rule and prompt v2, topics written in a minute use the generic prompt without the rule. This was found after the predeclared run (which expected F1 ≥ 0.80 and missed it with the frozen policy) and is reported as a diagnosis, not a tuned result. Coverage is 97–98 %: in 34 of 1,500 answers the generic prompt cited the affair ID instead of a passage ID, which the validator rejects rather than invent evidence.

5.4 Bias and neutrality

Frozen configuration, test split, decided affairs:

Slice	Tagged affairs	Precision	Recall
German / French / Italian (F1)	670 / 325 / 205 affairs		F1 0.90 / 0.85 / 0.85
Federal Assembly	195	0.84	0.87
Bern / Zurich / Geneva / Valais / Basel-Stadt	25 / 24 / 16 / 15 / 13	0.88 / 0.92 / 0.82 / 0.93 / 1.00	0.88 / 0.96 / 0.88 / 0.87 / 0.85
Title and text / title only	256 / 142	0.86 / 0.91	0.88 / 0.87
Federal Council bills	40	0.94	0.75

Title-only affairs, mostly cantonal, are now recalled as well as affairs with text; with the first prompt they lost 13 recall points (0.78 vs. 0.91, Bern 0.73, Basel-Stadt 0.62). The term rule finds German affairs slightly more often. Federal Council bills, with long legal titles where digitalisation is one aspect among many, are recalled least. Author and party are not in the affairs export; a party check is open. The prompt instructs Apertus to classify subject matter only and never judge a position.

5.5 Standard topics: how hard is APS?

Both team members labelled 20 affairs (12 federal, 8 cantonal) blind with an APS chapter and topic, reading French and Italian affairs in German. They agree on 11 of 20 chapters (55 %, κ 0.51) and 9 of 20 topics; most disagreements are multi-aspect affairs (*no transfer of tax data without anonymisation*: data protection or taxes). A single unadjudicated label would make model scoring ambiguous. This agreement is not a measured model accuracy ceiling; APS needs multi-label consensus labels before quality claims. The APS scheme (8 domains, 22 chapters, 165 topics) is exported to `configs/taxonomies/aps_topics.json`.

5.6 Cost and runtime

The frozen configuration needs 1,102 / 80 tokens per affair on average; the term rule decides 21 % of the test affairs without a model call. Measured throughput on the shared CSCS endpoint varied with load: the 1,200 test affairs took 379 s with 8 parallel requests at midday (about 150 answered requests per minute); a 500-affair probe with 16 parallel requests and no client limit reached 64 per minute in the late afternoon (p95 latency 71 s) without a single rate-limit error. Classifying every one of 325,000 affairs for one topic therefore takes about 28–67 hours on one key; the client's default limit of 30 requests per minute (`LLM_REQUESTS_PER_MINUTE`, an unverified quota assumption, see `docs/performance.md`) would stretch this to about 140 hours and can be raised or disabled.

At the Public AI list price for Apertus 70B (USD 0.82 / 2.92 per million input / output tokens) one topic over the full corpus costs about USD 370; on CSCS or a Swiss provider the price depends on the contract. Reading every affair for every new topic does not scale; sending only candidate affairs to Apertus is the next step.

5.7 Can a compiled query replace reading every affair? (negative result)

Idea: Apertus reads the topic once and writes search terms per language (`parltopic compile-topic`: 18–26 s, about 1,900 tokens, 40–56 terms per language; editable JSON); a cheap search returns candidates, and Apertus verifies only those (`parltopic find`). Measured on the full population behind E1, not on the balanced sample: every affair of the 35 PolitMonitor parliaments and the same affair types.

Period	Affairs	Tagged	Pre-filter	Candidates (share)	Tagged among candidates
2018–2023 (dev)	72,741	2,487	local, word-prefix, terms v1	4,703 (6.5 %)	61.0 %
2018–2023 (dev)	72,741	2,487	local, word-prefix, terms v2 (stems)	2,932 (4.0 %)	59.5 %
2024–2026 (test)	37,258	853	OpenParlData search, titles	6,073 (16.3 %)	59.1 %
2024–2026 (test)	37,258	853	local, word-prefix, terms v1	2,982 (8.0 %)	67.6 %

On the test sample, the frozen classifier restricted to the local candidates reaches F1 0.749 (precision 0.873, recall 0.656) against 0.876 when every affair is read. We set the bar before measuring (≥ 85 % of the tagged affairs among the candidates) and stopped: about 40 % of the tagged affairs contain no predictable search term ("Nevo/Rialto – Wie weiter?", "Handyfreie Schulen?!", smart glasses, robotaxis, a pension affair with a digital side aspect). High recall needs Apertus to read every affair, which is what the main configuration does. The compiler is still useful as a fast first pass — 60–68 % of the hits sit in 4–8 % of the affairs and arrive in minutes, the complete pass follows — and its terms are a readable, editable description of a topic. The demo's live search view uses it this way: the newest search hits per parliament, ranked by the compiled terms in the title, are checked first (experiment log, A · D). OpenParlData's own search would serve it if it supported prefixes (`digital:*` returns nothing, so compounds such as "Digitalisierungsstrategie" are missed) and date filters (`date_from`/`date_to` are ignored; deep offsets take about 60 s per page).

6. Limitations

- **Labels:** PolitMonitor tags are positives only; precision is a proxy, not a guaranteed bound. One topic is evaluated in depth; three more against official subject areas.
- **Test split:** opened validation, not a sealed holdout; a later, manually adjudicated holdout is needed for an independent quality claim.
- **Coverage:** municipal affairs are too few in the labels (6 in the test split); Romansh is supported by the schema but not evaluated.
- **Title-only affairs** are recalled worse; document text did not help significantly yet because long reports bring incidental mentions.
- **Standard topics:** the APS scheme is available, but a multi-label APS mode is not evaluated.
- **Full-corpus runtime:** high recall needs every affair read (28–67 h per topic over 325,000 affairs projected from historical higher-throughput runs, not a completed corpus measurement); the default limit of 30 requests per minute stretches this to about 140 h (5.6). The submission deliberately runs small functional checks only; a lexical pre-filter keeps only about 60 % of the hits (5.7).
- **Party bias** is not checked yet.
- **Topic policy:** the term rule and prompt v2 suit curated topics only; quickly written topics need the generic prompt (5.3), which in 2 % of answers cites an affair ID instead of a passage.

7. Reproducibility

```
python -m pip install -e '.[dev]'          # Python 3.11+; copy .env.example to .env, add LLM_API_KEY
make test                                  # unit tests, no API calls
parltopic fetch-results && parltopic claims # recompute every number in this report, no API calls
parltopic download-data                    # exports and label files to data/raw (~350 MB)
parltopic build-eval                       # rebuilds data/eval byte for byte (seed 42)
parltopic baseline --eval-set data/eval/e1_digitalisation_test.jsonl --topic configs/topics/digitalisation.yaml
parltopic run-eval --eval-set data/eval/e1_digitalisation_test.jsonl \
  --topic configs/topics/digitalisation.yaml --output results/test.jsonl --workers 8
parltopic compare-runs --eval-set … --baseline-results … --candidate-results …   # paired bootstrap
```

Every `run-eval` output has a manifest with the dataset hash, topic, system prompt, settings and source-file hashes; resuming with changed inputs is refused. Bounded submission uses a SQLite checkpoint journal, per-output writer lock and atomic JSONL snapshots; eight synthetic affairs exercise interrupted resume, including process termination. No 300k run was executed. Compiler, classifier and thinking share HTTP transport and process-local quota admission. All settings are in `.env.example`. `make run` starts the demo without an API key, with the bundled six-case historical replay and one recorded live search. Export preserves all evidence and checks the declared topic manifest; the isolated PostgreSQL check verifies version retention, repeated import and foreign keys.

8. Next steps

1. A semantic pre-filter (multilingual embeddings, e.g. bge-m3 hosted in Switzerland) measured against the 85 % recall bar that the lexical one missed; nightly runs only for new affairs.
2. Feed reviewer decisions (the demo stores confirm / reject; `affair_topics.review_decision`) back as labels and as in-scope / out-of-scope examples of the topic.
3. Multi-label APS chapters, evaluated against the human agreement in 5.5.
4. Party bias check; a sealed, adjudicated holdout including municipal affairs.

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